

Unit 3 – Exercises: Complete Step-by-Step Solutions

Data Distribution, Outliers, Skewness & Continuous Distributions

Exercise 1 — Mean, Median, Mode, Range

Dataset: 12, 15, 18, 15, 20, 22, 15, 25 (n = 8)

Step 1: Arrange the data in ascending order

12, 15, 15, 15, 18, 20, 22, 25

Step 2: Calculate the Mean

$$\text{Mean} = (12+15+18+15+20+22+15+25) / 8 = 142 / 8$$

Answer: Mean = 17.75

Step 3: Calculate the Median

With n = 8 (even), the median is the average of the 4th and 5th values in the sorted list.

$$\text{Median} = (15 + 18) / 2$$

Answer: Median = 16.5

Step 4: Calculate the Mode

The mode is the value that occurs most frequently. 15 occurs 3 times, more than any other value.

Answer: Mode = 15

Step 5: Calculate the Range

$$\text{Range} = \text{Maximum} - \text{Minimum} = 25 - 12$$

Answer: Range = 13

Exercise 2 — Frequency Distribution

Dataset: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4 (n = 12)

Step 1: Identify unique values

The values range from 2 to 6.

Step 2: Count the frequency of each value

Value	Frequency
2	3
3	3
4	3
5	2
6	1
Total	12

Step 3: Calculate relative frequency and percentage

For example, for value 2:

$$\text{Relative Frequency} = 3 / 12 = 0.25 \rightarrow \text{Percentage} = 25\%$$

Similarly, for the remaining values:

Value	Frequency	Percentage
2	3	25%
3	3	25%
4	3	25%
5	2	16.67%
6	1	8.33%

Conclusion

Answer: Values 2, 3 and 4 are the most common (25% each); 6 is the least common (8.33%).

Exercise 3 — Quartiles and IQR

Dataset: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30 (n = 10, already sorted)

Step 1: Find Q2 (the median)

With 10 observations, Q2 is the average of the 5th and 6th values.

$$Q2 = (15 + 18) / 2$$

Answer: Q2 = 16.5

Step 2: Find Q1

Lower half: 5, 8, 10, 12, 15 → middle value:

Answer: Q1 = 10

Step 3: Find Q3

Upper half: 18, 20, 22, 25, 30 → middle value:

Answer: Q3 = 22

Step 4: Calculate IQR

$$IQR = Q3 - Q1 = 22 - 10$$

Answer: IQR = 12

Exercise 4 — Outlier Detection (IQR Method)

Dataset: 10, 12, 13, 15, 16, 17, 18, 20, 45 (n = 9, already sorted)

Step 1: Find Q1

The median of 9 values is the 5th value (16), which is excluded from both halves.

Lower half: 10, 12, 13, 15 → $Q1 = (12 + 13) / 2$

Answer: Q1 = 12.5

Step 2: Find Q3

Upper half: 17, 18, 20, 45 $\rightarrow Q3 = (18 + 20) / 2$

Answer: $Q3 = 19$

Step 3: Calculate IQR

$$IQR = Q3 - Q1 = 19 - 12.5$$

Answer: $IQR = 6.5$

Step 4: Calculate the lower boundary

$$\text{Lower Bound} = Q1 - 1.5 \times IQR = 12.5 - 1.5(6.5) = 12.5 - 9.75$$

Answer: Lower Bound = 2.75

Step 5: Calculate the upper boundary

$$\text{Upper Bound} = Q3 + 1.5 \times IQR = 19 + 1.5(6.5) = 19 + 9.75$$

Answer: Upper Bound = 28.75

Step 6: Identify the outlier

Any value below 2.75 or above 28.75 is an outlier. Since $45 > 28.75$:

Answer: 45 is an outlier

Exercise 5 — Skewness (Asymmetry)

Dataset: 5, 6, 7, 7, 8, 8, 9, 20 (n = 8, already sorted)

Step 1: Find the Mean

$$\text{Mean} = (5+6+7+7+8+8+9+20) / 8 = 70 / 8$$

Answer: Mean = 8.75

Step 2: Find the Median

With n = 8, the median is the average of the 4th and 5th values.

$$\text{Median} = (7 + 8) / 2$$

Answer: Median = 7.5

Step 3: Compare Mean and Median

Mean = 8.75, Median = 7.5, so Mean > Median.

Interpretation

The large value 20 pulls the mean to the right of the median.

Answer: The distribution is positively (right) skewed

Exercise 6 — Continuous Uniform Distribution

A variable is uniformly distributed between 10 and 50. Find $P(20 < X < 35)$.

Step 1: Identify limits

$a = 10$, $b = 50$; we need $P(20 < X < 35)$, so $c = 20$, $d = 35$.

Step 2: Apply the uniform distribution formula

$$P(20 < X < 35) = (35 - 20) / (50 - 10) = 15 / 40$$

Answer: $P(20 < X < 35) = 0.375$, i.e. 37.5%

Exercise 7 — Z-Score for a Normal Distribution

A normal distribution has Mean = 100 and Standard Deviation = 15. Find the Z-score for $X = 130$.

Step 1: Formula

Step 2: Substitute values

$$Z = (130 - 100) / 15 = 30 / 15$$

Answer: $Z = 2$

Interpretation

$X = 130$ lies 2 standard deviations above the mean.

Exercise 8 — Normal Distribution Probability

Using the same distribution (Mean = 100, SD = 15), find $P(X < 130)$.

Step 1: Use the Z-score already found

From Exercise 7, $Z = 2$.

Step 2: Look up the standard normal table

$$P(Z < 2) \approx 0.9772$$

Step 3: State the probability

Answer: $P(X < 130) = 0.9772$, i.e. 97.72%

Interpretation

Approximately 97.72% of the values in this distribution are expected to fall below 130.

Exercise 9 — Comparing Mean and Median Before/After Outlier Treatment

Dataset: 10, 12, 13, 14, 15, 16, 17, 18, 20, 50 ($n = 10$, already sorted)

Step 1: Calculate Mean and Median before treatment

$$\text{Mean} = (10+12+13+14+15+16+17+18+20+50) / 10 = 185 / 10 = 18.5$$

$$\text{Median} = (15 + 16) / 2 = 15.5$$

Answer: Before treatment → Mean = 18.5, Median = 15.5 (gap = 3.0)

Step 2: Treat the outlier

50 is an extreme value (as confirmed by the IQR method in Exercise/Problem 4 style analysis). Replacing it with the median value 15.5 gives the treated dataset:

10, 12, 13, 14, 15, 15.5, 16, 17, 18, 20

Step 3: Calculate Mean and Median after treatment

$$\text{Mean} = (10+12+13+14+15+15.5+16+17+18+20) / 10 = 150.5 / 10 = 15.05$$

$$\text{Median} = (15 + 15.5) / 2 = 15.25$$

Answer: After treatment → Mean = 15.05, Median = 15.25 (gap = 0.20)

Interpretation

Before treatment the mean (18.5) is pulled well above the median (15.5) because of the extreme value 50, indicating skew caused by the outlier. After replacing the outlier with the median, the mean (15.05) and median (15.25) are very close, showing the distribution is now roughly symmetric and no longer distorted by an extreme value.

Exercise 10 — Complete EDA on a Student-Marks Dataset

Sample dataset (marks out of 100 for 10 students): 45, 50, 55, 60, 62, 65, 70, 72, 75, 95 (n = 10, already sorted)

A. Central Tendency

Step 1: Mean

$$\text{Mean} = (45+50+55+60+62+65+70+72+75+95) / 10 = 649 / 10$$

Answer: Mean = 64.9

Step 2: Median

$$\text{Median} = (62 + 65) / 2$$

Answer: Median = 63.5

Step 3: Mode

No value repeats in this dataset.

Answer: Mode = None (all values are unique)

B. Dispersion

Step 4: Range

$$\text{Range} = \text{Maximum} - \text{Minimum} = 95 - 45$$

Answer: Range = 50

Step 5: Quartiles and IQR

Lower half: 45, 50, 55, 60, 62 $\rightarrow Q1 = 55$

Upper half: 65, 70, 72, 75, 95 $\rightarrow Q3 = 72$

$$\text{IQR} = Q3 - Q1 = 72 - 55$$

Answer: IQR = 17

Step 6: Standard Deviation

Using the population standard deviation formula on the deviations from the mean (64.9):

$$\text{Variance} = \Sigma(X - \text{Mean})^2 / n \approx 1832.9 / 10 = 183.29$$

$$\text{SD} = \sqrt{183.29}$$

Answer: SD $\approx$ 13.54

C. Outlier Detection (IQR Method)

Step 7: Calculate boundaries

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR} = 55 - 1.5(17) = 55 - 25.5 = 29.5$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR} = 72 + 1.5(17) = 72 + 25.5 = 97.5$$

Since $45 \text{ (min)} > 29.5$ and $95 \text{ (max)} < 97.5$, both boundaries are satisfied.

Answer: No outliers detected by the $\pm 1.5 \times \text{IQR}$ rule (95 is high but still within the upper bound)

D. Skewness

Step 8: Compare Mean and Median

Mean = 64.9, Median = 63.5, so Mean > Median (slightly).

Answer: The distribution is mildly positively (right) skewed, due to the higher score of 95

E. Summary

Measure	Value
Mean	64.9
Median	63.5
Mode	None
Range	50
Q1 / Q3	55 / 72
IQR	17
Standard Deviation	≈ 13.54
Outliers (IQR rule)	None
Skewness	Mildly positive (right)

Overall Interpretation

The class performed moderately well, with marks clustered mainly between 55 and 75. There is a mild positive skew because one student (95) scored notably higher than the rest, though this value is not extreme enough to be classified as a statistical outlier.